TRIGONOMETRY

(त्रिकोणमिति)

SHEET-01

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

TRIGONOMETRY (त्रिकोणमिति)

CLASSROOM SHEET-01

Questions Based on Basic Trigonometric Ratios

1. If $\sin\theta = \frac{7}{25}$, then $\cos\theta$ and $\tan\theta$ are repsectively _____ .
(where θ is in the first quadrant)

यदि $\sin\theta = \frac{7}{25}$ है, तो $\cos\theta$ और $\tan\theta$ के मान क्रमशः _____ हैं।
(जहाँ θ प्रथम चतुर्थांश में है)

SSC CHSL 09/07/2024 (Shift-03)

(a) $\frac{25}{24}$ and $\frac{24}{7}$ (b) $\frac{25}{7}$ and $\frac{7}{24}$

(c) $\frac{24}{25}$ and $\frac{24}{7}$ (d) $\frac{24}{25}$ and $\frac{7}{24}$

2. What is the value of sec(t), if $\tan(t) = \frac{1}{3}$?

यदि $\tan(t) = \frac{1}{3}$ है, तो sec(t) का मान कितना है?

SSC CGL 09/09/2024 (Shift-02)

(a) $\frac{2\sqrt{2}}{3}$ (b) $\frac{\sqrt{10}}{9}$

(c) $\frac{\sqrt{10}}{3}$ (d) $\frac{\sqrt{3}}{3}$

3. If $\tan A = \frac{2}{5}$ find the value of $\frac{\sec^2 A}{\text{cosec}^2 A}$.

यदि $\tan A = \frac{2}{5}$ है, तो $\frac{\sec^2 A}{\text{cosec}^2 A}$ का मान ज्ञात कीजिए।

SSC CPO 03/10/2023 (Shift-3)

(a) $\frac{3}{5}$ (b) $\frac{4}{25}$

(c) $\frac{2}{5}$ (d) $\frac{9}{25}$

4. If $\cot A = \frac{5}{12}$, then find the value of $\frac{\sqrt{\sec A - \text{cosec} A}}{\sqrt{\sec A + \text{cosec} A}} - \sqrt{\frac{17}{7}}$.

यदि $\cot A = \frac{5}{12}$ है, तो $\frac{\sqrt{\sec A - \text{cosec} A}}{\sqrt{\sec A + \text{cosec} A}} - \sqrt{\frac{17}{7}}$ का मान ज्ञात कीजिए।

SSC CHSL 08/07/2024 (Shift-01)

(a) $\frac{24}{\sqrt{119}}$ (b) 0

(c) $-\frac{10}{\sqrt{119}}$ (d) $\frac{10}{\sqrt{119}}$

5. If $\cos\theta = \frac{\sqrt{3}}{2}$, then $\tan^2\theta \cos^2\theta$ = ?

यदि $\cos\theta = \frac{\sqrt{3}}{2}$ है, तो $\tan^2\theta \cos^2\theta$ = ?

SSC CGL 14/07/2023 (Shift-1)

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) $\sqrt{3}$

6. If $\tan\theta = \frac{7}{4}$, then the value of $\frac{4\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta}$ is equal to:

यदि $\tan\theta = \frac{7}{4}$ है, तो $\frac{4\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta}$ का मान ज्ञात कीजिए।

SSC CHSL 09/07/2024 (Shift-02)

(a) $\frac{2}{5}$ (b) $\frac{1}{3}$

(c) $\frac{4}{5}$ (d) $\frac{3}{5}$

7. If $\tan\theta = \frac{8}{15}$, and θ is an acute angle, then the value of $\frac{\sqrt{1-\sin\theta}}{\sqrt{1+\sin\theta}}$ is:

यदि $\tan\theta = \frac{8}{15}$ और θ कोई न्यूनकोण हो, तो $\frac{\sqrt{1-\sin\theta}}{\sqrt{1+\sin\theta}}$ का मान ज्ञात कीजिए।

SSC CGL 18/09/2024 (Shift-02)

(a) $\frac{2}{5}$ (b) $\frac{3}{5}$

(c) $\frac{1}{5}$ (d) $\frac{4}{5}$

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8. If $\sqrt{13}\ \sin\theta = 2$, then the value of $\frac{3\tan\theta + \sqrt{13}\sin\theta}{\sqrt{13}\cos\theta - 3\tan\theta}$ is:

यदि $\sqrt{13}\ \sin\theta = 2$ है, तो $\frac{3\tan\theta + \sqrt{13}\sin\theta}{\sqrt{13}\cos\theta - 3\tan\theta}$ का मान ज्ञात कीजिए।

SSC CHSL 02/07/2024 (Shift-04)

(a) 4 (b) 3
(c) 2 (d) 0

9. If $\sin A = \frac{2}{5}$, Where A is an acute angle, what is the value of $\frac{5\sin A + 2\,\text{cosec}\,A}{\sqrt{21}\sec A}$?

यदि $\sin A = \frac{2}{5}$ है, जहाँ A एक न्यूनकोण है, तो $\frac{5\sin A + 2\,\text{cosec}\,A}{\sqrt{21}\sec A}$ का मान क्या है?

SSC CHSL 10/08/2023 (Shift-01)

(a) $\frac{5}{4}$ (b) $\frac{7}{5}$
(c) $\frac{4}{5}$ (d) $\frac{5}{7}$

10. If $\sec A = \frac{17}{8}$, given that $A < 90°$, what is the value of the following? $\frac{34\sin A + 15\cot A}{68\cos A - 16\tan A}$

यदि $\sec A = \frac{17}{8}$ है, दिया है, $A < 90°$ तो निम्नलिखित का मान क्या है? $\frac{34\sin A + 15\cot A}{68\cos A - 16\tan A}$

SSC CGL 11/04/2022 (Shift- 03)

(a) 23 (b) 19
(c) 30 (d) 38

11. If $7\tan\theta = 3$, then $\frac{7\sin\theta + 5\cos\theta}{7\sin\theta - 2\cos\theta}$ is:

यदि $7\tan\theta = 3$ है, तो $\frac{7\sin\theta + 5\cos\theta}{7\sin\theta - 2\cos\theta}$ का मान ज्ञात कीजिए।

SSC CGL 12/09/2024 (Shift-02)

(a) 7 (b) 5
(c) 8 (d) 6

12. Given that $\sin\theta = \frac{15}{17}$. Evaluate $\frac{15\cot\theta + 17\sin\theta}{8\tan\theta + 16\sec\theta}$, where $0° < \theta < 90°$.

$\sin\theta = \frac{15}{17}$ दिया गया है। $\frac{15\cot\theta + 17\sin\theta}{8\tan\theta + 16\sec\theta}$ का मान ज्ञात कीजिए, जहां $0° < \theta < 90°$ है।

SSC CHSL 02/07/2024 (Shift-02)

(a) $\frac{49}{26}$ (b) $\frac{23}{49}$
(c) $\frac{49}{23}$ (d) $\frac{26}{23}$

13. If $\cos\theta = \frac{12}{13}$, then the value of $\frac{\sin\theta(1-\tan\theta)}{\tan\theta(1+\text{cosec}\theta)}$ is :

यदि $\cos\theta = \frac{12}{13}$ है तो $\frac{\sin\theta(1-\tan\theta)}{\tan\theta(1+\text{cosec}\theta)}$ का मान ज्ञात कीजिए।

SSC CGL MAINS 03/02/2022

(a) $\frac{25}{78}$ (b) $\frac{35}{234}$
(c) $\frac{35}{108}$ (d) $\frac{25}{156}$

14. If $\left\{\frac{(3\sin\theta - \cos\theta)}{(\cos\theta + \sin\theta)}\right\} = 1$, then the value of $\cot\theta$ is:

यदि $\left\{\frac{(3\sin\theta - \cos\theta)}{(\cos\theta + \sin\theta)}\right\} = 1$ है, तो $\cot\theta$ का मान ज्ञात कीजिए।

SSC CGL 14/07/2023 (Shift-1)

(a) 3 (b) 0
(c) 1 (d) 2

15. If $\frac{7\sin\theta + 4\cos\theta}{9\sin\theta - 2\cos\theta} = \frac{5}{4}$, then the value of $\frac{\tan^2\theta + 5}{\tan^2\theta - 5}$ is:

यदि $\frac{7\sin\theta + 4\cos\theta}{9\sin\theta - 2\cos\theta} = \frac{5}{4}$ है, तो $\frac{\tan^2\theta + 5}{\tan^2\theta - 5}$ किसके बराबर होगा?

SSC CGL 09/09/2024 (Shift-03)

(a) $\frac{2121}{769}$ (b) $-\frac{2121}{769}$
(c) $\frac{769}{2121}$ (d) $-\frac{769}{2121}$

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16. If $\sin\theta = \frac{12}{13}$, $0° < \theta < 90°$, then $\frac{\sin^2\theta - \cos^2\theta}{2\sin\theta.\cos\theta} \times \frac{1}{\tan^2\theta} =$ _____.

यदि $\sin\theta = \frac{12}{13}$, $0° < \theta < 90°$ है तो $\frac{\sin^2\theta - \cos^2\theta}{2\sin\theta.\cos\theta} \times \frac{1}{\tan^2\theta} =$ _____.

SSC CHSL 08/06/2022 (Shift- 2)

(a) $\frac{295}{3456}$ (b) $\frac{290}{3542}$

(c) $\frac{695}{3542}$ (d) $\frac{595}{3456}$

17. If $\tan\theta = \frac{2}{\sqrt{11}}$, $0° < \theta < 90°$, then the value of $\frac{2\text{cosec}^2\theta - 3\sec^2\theta}{3\text{cosec}^2\theta + 4\sec^2\theta}$ is equal to :

यदि $\tan\theta = \frac{2}{\sqrt{11}}$, $0° < \theta < 90°$ है तो $\frac{2\text{cosec}^2\theta - 3\sec^2\theta}{3\text{cosec}^2\theta + 4\sec^2\theta}$ का मान ज्ञात कीजिए।

CHSL 26/10/2020 (Shift- 03)

(a) $\frac{11}{45}$ (b) $\frac{11}{49}$

(c) $\frac{13}{49}$ (d) $\frac{10}{49}$

18. If $\sin\theta = \frac{2xy}{x^2+y^2}$, then the value of $\tan\theta$ is:

यदि $\sin\theta = \frac{2xy}{x^2+y^2}$ तो $\tan\theta$ का मान है:

SSC CHSL 01/07/2024 (Shift-02)

(a) $\frac{2xy}{x^2-y^2}$ (b) $\frac{x^2+y^2}{x^2-y^2}$

(c) $\frac{x^2-y^2}{2xy}$ (d) $\frac{x^2-y^2}{x^2+y^2}$

19. If $\sec\theta = \frac{a}{b}$, $b \neq 0$, then $\frac{1-\tan^2\theta}{2-\sin^2\theta} = ?$

यदि $\sec\theta = \frac{a}{b}$, $b \neq 0$ है तो $\frac{1-\tan^2\theta}{2-\sin^2\theta} = ?$

SSC CGL MAINS (15/10/2020)

(a) $\frac{a^2(2b^2+a^2)}{b^2(a^2-b^2)}$ (b) $\frac{a^2(2b^2+a^2)}{b^2(a^2+b^2)}$

(c) $\frac{a^2(2b^2-a^2)}{b^2(a^2+b^2)}$ (d) $\frac{a^2(2b^2-a^2)}{a^2(a^2+b^2)}$

20. If $\cot A = 7$, then the value of $\frac{5\cos A + 4\sin A}{\cos^3 A + 7\sin^3 A + 6\sin A}$ is:

यदि $\cot A = 7$ है, तो $\frac{5\cos A + 4\sin A}{\cos^3 A + 7\sin^3 A + 6\sin A}$ का मान ज्ञात करें।

SSC CHSL 01/07/2024 (Shift-03)

(a) 3 (b) 2

(c) 0 (d) 1

21. If $\frac{\text{cosec}\,\theta + \cot\theta}{\text{cosec}\,\theta - \cot\theta} = 7$, then the value of $\frac{4\sin^2\theta - 1}{4\sin^2\theta + 5}$ is:

यदि $\frac{\text{cosec}\,\theta + \cot\theta}{\text{cosec}\,\theta - \cot\theta} = 7$ है तो $\frac{4\sin^2\theta - 1}{4\sin^2\theta + 5}$ का मान ज्ञात कीजिए।

SSC CGL 16/08/2021 (Shift- 01)

(a) $\frac{1}{3}$ (b) $-\frac{1}{3}$

(c) $-\frac{1}{9}$ (d) $\frac{1}{9}$

22. If $\frac{\sin^2\theta}{\tan^2\theta - \sin^2\theta} = 5$, θ is an acute angle, then the value of $\frac{24\sin^2\theta - 15\sec^2\theta}{6\text{cosec}^2\theta - 7\cot^2\theta}$ is:

यदि $\frac{\sin^2\theta}{\tan^2\theta - \sin^2\theta} = 5$, θ न्यूनकोण है तो $\frac{24\sin^2\theta - 15\sec^2\theta}{6\text{cosec}^2\theta - 7\cot^2\theta}$ का मान ज्ञात कीजिए।

SSC CGL 23/08/2021 (Shift- 02)

(a) 2 (b) –14

(c) 14 (d) –2

23. If $\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta} = \frac{1}{7}$, θ lies in first quadrant, then the value of $\frac{\text{cosec}\,\theta + \cot^2\theta}{\text{cosec}\,\theta - \cot^2\theta}$ is:

यदि $\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta} = \frac{1}{7}$, θ प्रथम चतुर्थांश में स्थित है तो $\frac{\text{cosec}\,\theta + \cot^2\theta}{\text{cosec}\,\theta - \cot^2\theta}$ का मान है।

SSC CGL MAINS 03/02/2022

(a) $\frac{19}{5}$ (b) $\frac{22}{3}$

(c) $\frac{37}{12}$ (d) $\frac{37}{19}$

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24. In a ΔABC, right angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, then sinA. cosC + cosA. sinC = _____.

एक ΔABC में, B पर समकोण है, यदि $\tan A = \frac{1}{\sqrt{3}}$ है, तो sinA. cosC + cosA. sinC का मान ज्ञात कीजिए।

SSC CHSL 26/05/2022 (Shift- 1)

(a) 0 (b) 2
(c) – 1 (d) 1

25. In ΔABC, right angled at B, if $\tan A = \frac{1}{2}$, then the value of $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$ is :

एक ΔABC में, B पर समकोण है, यदि $\tan A = \frac{1}{2}$ है, तो $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$ का मान ज्ञात कीजिए।

SSC CGL TIER-II (16/10/2020)

(a) $2\sqrt{5}$ (b) 3
(c) 2 (d) 1

26. In ΔABC, ∠B = 90° and AB : BC = 1 : 2. The value of cos A + tan C is:

ΔABC में, ∠B = 90° और AB : BC = 1 : 2 है। cos A + tan C का मान_____ है।

SSC CPO 04/10/2023 (Shift-01)

(a) $\frac{1+\sqrt{5}}{2\sqrt{5}}$ (b) $\frac{2\sqrt{5}}{2+\sqrt{5}}$
(c) $\frac{5+\sqrt{5}}{2\sqrt{5}}$ (d) $\frac{2+\sqrt{5}}{2\sqrt{5}}$

27. If $\sin A = \frac{5}{13}$ and 7cot B = 24, then the value of (secA cosB)(cosecBtan A) is:

यदि $\sin A = \frac{5}{13}$ तथा 7cot B = 24 है तो (secA cosB)(cosecBtan A) का मान ज्ञात कीजिए।

SSC CGL MAINS/29/01/2022

(a) $\frac{13}{7}$ (b) $\frac{65}{42}$
(c) $\frac{15}{13}$ (d) $\frac{13}{14}$

28. If $\sin 23° = \frac{a}{b}$, then the value of sec23° – sin67° is _____.

यदि $\sin 23° = \frac{a}{b}$ है, तो sec23° – sin67° का मान क्या है?

SSC CGL 14/07/2023 (Shift-3)

(a) $\frac{a^2}{\sqrt{b^2 - a^2}}$ (b) $\frac{b^2 - a^2}{ab}$
(c) $\frac{a^2}{b\sqrt{b^2 + a^2}}$ (d) $\frac{a^2}{b\sqrt{b^2 - a^2}}$

29. If sin31° = α, then the value of cot59° is:

यदि sin31° = α है, तो cot59° का मान ज्ञात कीजिए।

SSC CGL 13/09/2024 (Shift-01)

(a) $\frac{\sqrt{1-\alpha^2}}{2\alpha}$ (b) $\frac{\alpha}{\sqrt{1-\alpha^2}}$
(c) $\frac{\sqrt{1+\alpha^2}}{\alpha}$ (d) $\frac{\alpha}{\sqrt{1+\alpha^2}}$

30. If cos 24° = $\frac{m}{n}$, then the value of (cosec24° – cos 66°) is:

यदि cos 24° = $\frac{m}{n}$ है, तो (cosec24° – cos 66°) का मान क्या होगा?

SSC CGL 17/09/2024 (Shift-01)

(a) $\frac{m^2}{n\sqrt{m^2 - n^2}}$ (b) $\frac{n^2}{n\sqrt{m^2 - n^2}}$
(c) $\frac{n^2}{m\sqrt{n^2 - m^2}}$ (d) $\frac{m^2}{n\sqrt{n^2 - m^2}}$

31. If $\sin\theta - \cos\theta = \frac{7}{13}$, $0° < \theta < 90°$, then the value of $\sin\theta + \cos\theta$ is :

यदि $\sin\theta - \cos\theta = \frac{7}{13}$, $0° < \theta < 90°$ है तो $\sin\theta + \cos\theta$ का मान ज्ञात कीजिए।

(a) $\frac{17}{13}$ (b) $\frac{13}{17}$
(c) $\frac{1}{13}$ (d) $\frac{1}{17}$

32. If $0° < A, B < 45°$, $\cos(A+B) = \frac{24}{25}$ and $\sin(A-B) = \frac{15}{17}$, then tan2A = ?

यदि $0° < A, B < 45°$, $\cos(A+B) = \frac{24}{25}$ तथा $\sin(A-B) = \frac{15}{17}$ है तो tan2A का मान ज्ञात कीजिए।

SSC CGL 06/03/2020 (Shift- 02)

(a) 0 (b) 1
(c) $\frac{416}{87}$ (d) $\frac{213}{4}$

Maths by Aditya Ranjan sir

Questions Based on Values of Trigonometric Ratios

33. Simplify $\frac{\sin 60° - \cos 60°}{\sin 60° + \cos 60°}$

$\frac{\sin 60° - \cos 60°}{\sin 60° + \cos 60°}$ को सरल कीजिए।

SSC CHSL 11/07/2024 (Shift-03)

(a) $2+\sqrt{3}$ (b) $\sqrt{3}-1$

(c) $2-\sqrt{3}$ (d) $\sqrt{3}-2$

34. What is the value of sin30° + cos30° + tan30°

sin30° + cos30° + tan30° का मान क्या है?

SSC CGL 14/07/2023 (Shift-3)

(a) $\frac{5+\sqrt{3}}{2\sqrt{3}}$ (b) $\frac{5-\sqrt{3}}{2\sqrt{3}}$

(c) $\frac{5-\sqrt{3}}{\sqrt{3}}$ (d) $\frac{5-\sqrt{3}}{\sqrt{3}}$

35. The value of $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$ at $\theta = 45°$ is:

$\theta = 45°$ पर $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$ का मान ज्ञात कीजिए।

SSC CGL 23/09/2024 (Shift-03)

(a) 1 (b) 0

(c) 3 (d) 2

36. If $\sqrt{3}\tan\theta = 3\sin\theta$, then what is the value of $\sin^2\theta - \cos^2\theta$?

यदि $\sqrt{3}\tan\theta = 3\sin\theta$ है तो $\sin^2\theta - \cos^2\theta$ का मान ज्ञात कीजिए।

SSC CGL MAINS 08/08/2022

(a) $\frac{1}{2}$ (b) $\frac{1}{5}$

(c) $\frac{1}{3}$ (d) $\frac{1}{4}$

37. If $2k\sin 30°\cos 30°\cot 60° = \frac{\cot^2 30°\sec 60°\tan 45°}{\text{cosec}^2 45°\text{cosec}30°}$, then find the value of k.

यदि $2k\sin 30°\cos 30°\cot 60° = \frac{\cot^2 30°\sec 60°\tan 45°}{\text{cosec}^2 45°\text{cosec}30°}$ है तो k का मान ज्ञात कीजिए।

SSC CGL 12/04/2022 (Shift- 03)

(a) $\frac{3}{2}$ (b) 3

(c) 1 (d) 6

38. If A = 10°, what is the value of $\frac{12\sin 3A + 5\cos(5A-5)°}{9\sin\frac{9A}{2} - 4\cos(5A+10)°}$

यदि A = 10° है तो $\frac{12\sin 3A + 5\cos(5A-5)°}{9\sin\frac{9A}{2} - 4\cos(5A+10)°}$ का मान ज्ञात कीजिए।

SSC CGL 18/04/2022 (Shift- 03)

(a) $\frac{6\sqrt{2}+5}{(9+2\sqrt{2})}$ (b) $\frac{6\sqrt{2}-5}{(9-2\sqrt{2})}$

(c) $\frac{6\sqrt{2}+5}{(9-2\sqrt{2})}$ (d) $\frac{(9-2\sqrt{2})}{6\sqrt{2}+5}$

39. If $3\tan\theta = 2\sqrt{3}\sin\theta$, $0° < \theta < 90°$, then the value of $\frac{\text{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2\theta + \tan^2 2\theta}$ is:

यदि $3\tan\theta = 2\sqrt{3}\sin\theta$, $0° < \theta < 90°$ है तो $\frac{\text{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2\theta + \tan^2 2\theta}$ का मान ज्ञात कीजिए।

SSC CGL MAINS 29/01/2022

(a) $\frac{4}{13}$ (b) $\frac{20}{39}$

(c) $\frac{4}{3}$ (d) $\frac{20}{27}$

40. If $4\sin^2(2x-10)° = 3, 0° \le (2x-10) \le 90°$, then find the value of $\frac{\sin^4(x-5)° + \cos^4(x-5)°}{1-2\sin^2(3x-15)°\cos^2(3x-15)°}$.

यदि $4\sin^2(2x-10)° = 3, 0° \le (2x-10) \le 90°$ है तो $\frac{\sin^4(x-5)° + \cos^4(x-5)°}{1-2\sin^2(3x-15)°\cos^2(3x-15)°}$ का मान ज्ञात कीजिए।

SSC CGL 17/08/2021 (Shift- 03)

(a) 1 (b) $\frac{5}{8}$

(c) $-\frac{5}{8}$ (d) -1

41. If $2\sin(3x-15)° = 1, 0° < (3x-15) < 90°$, then find the value of $\cos^2(2x+15)° + \cot^2(x+15)°$.

यदि $2\sin(3x-15)° = 1, 0° < (3x-15) < 90°$ है तो $\cos^2(2x+15)° + \cot^2(x+15)°$ का मान ज्ञात कीजिए।

SSC CGL 16/08/2021 (Shift- 02)

(a) 1 (b) $\frac{5}{2}$

(c) $-\frac{7}{2}$ (d) $\frac{7}{2}$

Maths by Aditya Ranjan sir

42. If (2 cos A + 1) (2 cos A – 1) = 0, 0° < A ? 90°, then find the value of A.

यदि (2 cos A + 1) (2 cos A – 1) = 0, 0° < A ? 90° है तो A का मान ज्ञात कीजिए।

SSC CGL 21/04/2022 (Shift- 01)

(a) 90° (b) 45°
(c) 30° (d) 60°

43. If $\sin (x + 30)° = \frac{3}{\sqrt{12}}$, then the value of x $(0° < x < 90°)$ is :

यदि $\sin (x + 30)° = \frac{3}{\sqrt{12}}$ है तो x $(0° < x < 90°)$ का मान ज्ञात कीजिए।

SSC CHSL 26/10/2020 (Shift- 02)

(a) 60° (b) 15°
(c) 45° (d) 30°

44. If $0° < \theta < 90°$ and $\cos^2\theta = 3 (\cot^2\theta - \cos^2\theta)$, then the value of $\left(\frac{1}{2}\sec\theta + \sin\theta\right)^{-1}$ is :

यदि $0° < \theta < 90°$ तथा $\cos^2\theta = 3 (\cot^2\theta - \cos^2\theta)$ है तो $\left(\frac{1}{2}\sec\theta + \sin\theta\right)^{-1}$ का मान ज्ञात कीजिए।

SSC CGL 04/06/2019 (Shift- 02)

(a) $\sqrt{3}+2$ (b) $2(2-\sqrt{3})$
(c) $2(\sqrt{3}-1)$ (d) $\sqrt{3}+1$

45. If $\frac{\cos^2\theta}{\cot^2\theta + \sin^2\theta - 1} = 3$, $0° < \theta < 90°$, then the value of $(\tan\theta + \operatorname{cosec}\theta)$ is:

यदि $\frac{\cos^2\theta}{\cot^2\theta + \sin^2\theta - 1} = 3$, $0° < \theta < 90°$ है तो $(\tan\theta + \operatorname{cosec}\theta)$ का मान ज्ञात कीजिए।

SSC CGL 13/08/2021 (Shift- 01)

(a) $2\sqrt{3}$ (b) $\frac{5\sqrt{3}}{3}$
(c) $3\sqrt{3}$ (d) $\frac{4\sqrt{3}}{3}$

46. If $\cos (A-B) = \frac{\sqrt{3}}{2}$ and $\sec A = 2$, $0° \le A \le 90°$, $0° \le B \le 90°$, then what is the measure of B?

यदि $\cos (A-B) = \frac{\sqrt{3}}{2}$ तथा $\sec A = 2$, $0° \le A \le 90°$, $0° \le B \le 90°$ है तो कोण B का मान ज्ञात कीजिए।

SSC CGL 11/04/2022 (Shift- 02)

(a) 60° (b) 0°
(c) 30° (d) 90°

47. If $\sin\left(\frac{2A+B}{2}\right) = \cos\left(\frac{2A-B}{2}\right) = \frac{\sqrt{3}}{2}$ and $0° < \frac{2A+B}{2} < 90°$ then find the value of sin[3(A – B)].

यदि $\sin\left(\frac{2A+B}{2}\right) = \cos\left(\frac{2A-B}{2}\right) = \frac{\sqrt{3}}{2}$ तथा $0° < \frac{2A+B}{2} < 90°$ है तो sin[3(A – B)] का मान ज्ञात कीजिए।

SSC CGL 16/08/2021 (Shift- 03)

(a) 1 (b) $\frac{1}{\sqrt{2}}$
(c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}2$

48. If triangle ABC is a right isosceles triangle, right angled at B, then find the value of $\frac{\sin(A-C)+\sin(A+C)-2\sin B}{\cot A+\cot B+\cot C}$.

यदि त्रिभुज ABC एक समकोण समद्विबाहु त्रिभुज है, जो B पर समकोण है, तो $\frac{\sin(A-C)+\sin(A+C)-2\sin B}{\cot A+\cot B+\cot C}$ का मान ज्ञात कीजिए।

NTPC CBT-1 08/04/2021 (Shift-03)

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$
(c) $\frac{3}{2}$ (d) 0

49. If $\tan^2 A - 6\tan A + 9 = 0$, $0° < A < 90°$, What is the value of $6\cot A + 8\sqrt{10}\cos A$?

यदि $\tan^2 A - 6\tan A + 9 = 0$, $0° < A < 90°$ है तो $6\cot A + 8\sqrt{10}\cos A$ का मान ज्ञात कीजिए।

SSC CGL 20/04/2022 (Shift- 02)

(a) $10\sqrt{10}$ (b) $8\sqrt{10}$
(c) 10 (d) 14

50. If $3\sec\theta + 4\cos\theta - 4\sqrt{3} = 0$ where θ is an acute angle then the value of θ is:

यदि $3\sec\theta + 4\cos\theta - 4\sqrt{3} = 0$ जहां θ एक न्यूनकोण है तो θ का मान ज्ञात कीजिए।

SSC CGL 13/08/2021 (Shift- 03)

(a) 20° (b) 30°
(c) 60° (d) 45°

51. Sin (– A) = ?

SSC CHSL 24/05/2022 (Shift- 02)

(a) cos A (b) – cos A
(c) – sin A (d) sin A

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52. tan (π + θ) = ?

SSC CHSL 08/06/2022 (Shift-01)

(a) secθ (b) cosecθ
(c) cotθ (d) tanθ

53. Find the exact value of sin150°.
sin150° का सटीक मान ज्ञात कीजिए।

SSC CGL 12/09/2024 (Shift-03)

(a) 1.5 (b) 0.5
(c) 0.75 (d) –0.5

54. What is the value of tan 570°
tan 570° का मान कितना होगा?

SSC CHSL 04/07/2024 (Shift-02)

(a) $-\frac{1}{\sqrt{3}}$ (b) $-\sqrt{3}$
(c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

55. What is the value of sin(–405°)?
sin(–405°) का मान ज्ञात करें।

SSC SSC 20/07/2023 (Shift-03)

(a) $\frac{\sqrt{5}}{2}$ (b) $\frac{1}{2}$
(c) $\frac{-1}{2}$ (d) $\frac{-1}{\sqrt{2}}$

56. Find the value of tan(–1125°).
tan(–1125°) का मान ज्ञात कीजिए।

SSC CGL 25/07/2023 (Shift-2)

(a) 1 (b) $\frac{1}{2}$
(c) –1 (d) 0

57. Find the value of tan4384° + cot6814° = ?
tan4384° + cot6814° का मान ज्ञात कीजिए।

SSC CGL 26/07/2023 (Shift-3)

(a) –1 (b) 2
(c) 0 (d) 1

58. Find the value of
cos570° sin510° + sin(–330°) cos(–390°)
का मान ज्ञात कीजिए।

(a) 1 (b) 0
(c) 2 (d) NOT

59. Find is the value of $\sin\frac{7\pi}{4}\ \sin\frac{\pi}{4}\ \sin\frac{3\pi}{4}\ \sin\frac{5\pi}{4}$.

$\sin\frac{7\pi}{4}\ \sin\frac{\pi}{4}\ \sin\frac{3\pi}{4}\ \sin\frac{5\pi}{4}$ का मान ज्ञात करें।

RRB JE 22/05/2019 (Shift-02)

(a) $\frac{1}{4}$ (b) $\frac{3}{16}$
(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

60. If $\tan(90° - \theta) = \frac{2}{\sqrt{3}}$, then the value of $2\sqrt{3}\tan\theta + 1$ is:

यदि $\tan(90° - \theta) = \frac{2}{\sqrt{3}}$, तो $2\sqrt{3}\tan\theta + 1$ का मान ज्ञात कीजिए।

SSC CHSL 11/08/2023 (Shift-2)

(a) 4 (b) 5
(c) 3 (d) 6

61. What is the simplified value of $\cos^2(90°-\theta) - \left[\frac{\{\cos(90° - \theta)\cos\theta\}}{\cot\theta}\right]$?/का सरलीकृत मान क्या है?

SSC CHSL 31/05/2022 (Shift- 3)

(a) 4 (b) 2
(c) 0 (d) 1

62. Find the value of

$$\frac{\sin(90° - \theta)\cos(90° - \theta)\cot(90° - \theta)}{\cos^2\theta - 1}$$

का मान ज्ञात कीजिए।

Group D 19/09/2022 (Shift-01)

(a) tanθ (b) 0
(c) – 1 (d) 2sinθ cosθ

63. The following expression is equal to/नीचे दिया गया व्यंजक बराबर है।
cot85° + cos75°

SSC CGL 20/07/2023 (Shift- 02)

(a) tan85° + sin75° (b) tan85° – sin75°
(c) tan5° + sin15° (d) tan5° – sin15°

64. The value of sin 73° + cos137° is
sin73° + cos137° का मान क्या होगा?

SSC CHSL 10/06/2022 (Shift- 3)

(a) sin 13° (b) cos 13°
(c) cos 18° (d) sin 18°

65. Find θ, if $\cos\theta = -\frac{\sqrt{3}}{2}$

यदि $\cos\theta = -\frac{\sqrt{3}}{2}$ है, तो θ का मान ज्ञात करें।

SSC CHSL 02/06/2022 (Shift- 2)

(a) $\frac{3\pi}{2}$ (b) $\frac{5\pi}{2}$
(c) $\frac{5\pi}{6}$ (d) $\frac{4\pi}{3}$

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66. The value of $\sin^2\frac{2\pi}{3}+\cos^2\frac{5\pi}{6}-\tan^2\frac{3\pi}{4}$ is:

$\sin^2\frac{2\pi}{3}+\cos^2\frac{5\pi}{6}-\tan^2\frac{3\pi}{4}$ का मान क्या है?

SSC Phase X 03/08/2022 (Shift- 02)

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) 4 (d) 2

67. Simplify. $\frac{1}{\cos x}\sqrt{\frac{\cos(\pi+x)\cos(-x)}{\sin(\pi-x)\cos\left(\frac{\pi}{2}+x\right)}}$

$\frac{1}{\cos x}\sqrt{\frac{\cos(\pi+x)\cos(-x)}{\sin(\pi-x)\cos\left(\frac{\pi}{2}+x\right)}}$ सरल करें

SSC Phase X 04/08/2022 (Shift- 03)

(a) $\tan x$ (b) $\cot x$
(c) $\sec x$ (d) $\operatorname{cosec} x$

68. If $\cos x = -\frac{\sqrt{3}}{2}$ and $\pi < x < \frac{3\pi}{2}$, then the value of $2\cot^2 x + 3\sec^2 x$ is :

SSC CHSL 08/07/2019 (Shift- 02)

(a) 10 (b) 4
(c) 8 (d) 16

69. Find x if $\cos x = -\frac{1}{2}$, $\left(\frac{\pi}{2} < x < \pi\right)$

SSC CHSL 15/10/2020 (Shift-03)

(a) $\frac{3\pi}{2}$ (b) $\frac{2\pi}{3}$
(c) $\frac{5\pi}{6}$ (d) $\frac{4\pi}{3}$

70. If $\cos x = -\frac{1}{2}$, x lies in third quadrant, then $\tan x = ?$

यदि $\cos x = -\frac{1}{2}$, x तीसरे चतुर्थांश में स्थित है, तो $\tan x = ?$

SSC CGL 17/07/2023 (Shift-03)

(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{3}}$

71. Find the value of the following?

$\cos^2(270^\circ-\phi)-\sin^2(180^\circ-\phi)+\sin^2\left(\frac{\pi}{2}\right)\sin^2(270^\circ-\phi)$.

निम्न का मान ज्ञात कीजिए?

RRB NTPC 08/01/2021 (Shift-01)

(a) $\cos^2\phi$ (b) $\sin^2\left(\frac{\pi}{2}\right)$
(c) $\sin^2(\phi)-1$ (d) $\sin^2(\phi)$

72. What is the value of $[\cos(90^\circ+A)\div\sec(270^\circ-A)]+[\sin(270^\circ+A)\div\operatorname{cosec}(630^\circ-A)]$

का मान ज्ञात कीजिए।

(a) 3 sec A (b) tanA secA
(c) 0 (d) 1

73. If $\sec\theta+\tan\theta = 2+\sqrt{5}$, then the value of $\sin\theta+\cos\theta$ is :

यदि $\sec\theta+\tan\theta = 2+\sqrt{5}$ है तो $\sin\theta+\cos\theta$ का मान ज्ञात कीजिए।

(a) $\frac{3}{\sqrt{5}}$ (b) $\sqrt{5}$
(c) $\frac{7}{\sqrt{5}}$ (d) $\frac{1}{\sqrt{5}}$

ANSWER KEY

1.(d)	2.(c)	3.(b)	4.(c)	5.(b)	6.(a)	7.(b)	8.(a)	9.(b)	10.(b)
11.(c)	12.(a)	13.(b)	14.(c)	15.(b)	16.(d)	17.(b)	18.(a)	19.(c)	20.(a)
21.(d)	22.(b)	23.(a)	24.(d)	25.(b)	26.(d)	27.(b)	28.(d)	29.(b)	30.(d)
31.(a)	32.(c)	33.(c)	34.(a)	35.(b)	36.(c)	37.(b)	38.(c)	39.(b)	40.(b)
41.(d)	42.(d)	43.(d)	44.(b)	45.(b)	46.(c)	47.(b)	48.(b)	49.(c)	50.(b)
51.(c)	52.(d)	53.(b)	54.(c)	55.(d)	56.(c)	57.(c)	58.(b)	59.(a)	60.(a)
61.(c)	62.(c)	63.(c)	64.(a)	65.(c)	66.(a)	67.(d)	68.(a)	69.(b)	70.(a)
71.(a)	72.(d)	73.(a)							

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CALCULATION TECHNIQUES

TECHNIQUE - 01

♦ Cube Root of Non-perfect cube ♦

$$\sqrt[3]{N} = A \pm \frac{B}{3.A^2}$$

First find a number which is a perfect cube near to the number N, let it be A and the difference between N and A^3 is B.

Note: '+' sign is used if $N > A^3$ & '–' sign is used if $N < A^3$

Ex. $\sqrt[3]{60} = 4 - \frac{4}{3 \times 4^2} = \frac{188}{48} = 3.92$

Ex. $\sqrt[3]{72} = 4 + \frac{8}{3 \times 4^2} = \frac{200}{48} = 4.17$

♦ Do it Yourself ♦

(i) $\sqrt[3]{10}$ = ...

(ii) $\sqrt[3]{66}$ = ...

(iii) $\sqrt[3]{118}$ = ...

(iv) $\sqrt[3]{226}$ = ...

(v) $\sqrt[3]{525}$ = ...

(vi) $\sqrt[3]{1365}$ = ...

(vii) $\sqrt[3]{1500}$ = ...

TECHNIQUE - 02

♦ Square of two digit number ♦

+ Carry + Carry

$(ab)^2 = a^2/2ab/b^2$

1 digit, 1 digit

- **Square of 14**

Here, a = 1 & b = 4

$14^2 = 1^2/2.1.4/4^2 = 1/8/16 = 196$

- **Square of 23**

Here, a = 2 & b = 3

$23^2 = 2^2/2.2.3/3^2 = 4/12/9 = 529$

♦ Do it Yourself ♦

(i) 18^2 = ... (ii) 23^2 = ...

(iii) 27^2 = ... (iv) 34^2 = ...

(v) 38^2 = ... (vi) 43^2 = ...

(vii) 57^2 = ... (viii) 64^2 = ...

(ix) 66^2 = ... (x) 75^2 = ...

(xi) 78^2 = ... (xii) 84^2 = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
TRIGONOMETRY SHEET - 01						
QUESTIONS BASED ON BASIC TRIGONOMETRIC RATIOS						
QUESTIONS BASED ON VALUES OF TRIGONOMETRIC RATIOS						